

Machine Learning Regression Assignment

Problem Statement

1.) Identify your problem statement

Model Name: IPP (Insurance Price Prediction)

Stage 1 = Machine Learning

Stage 2 = Supervised Learning

Stage 3 = Regression (Numerical value)

2.) Tell basic info about the dataset (Total number of rows, columns)

Columns :

age sex bmi children smoker charges

Rows :

1338

Inputs : age, sex, bmi, children, smoker

independent=dataset[['age', 'bmi', 'children', 'sex_male', 'smoker_yes']]

Output : charges

dependent=dataset[["charges"]]

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Smoker → yes → true

Smoker → no → false

Sex → Female → False

Sex → Male → True

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Refer the below tables

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Refer the below.

6.) Mention your final model, justify why u have chosen the same.

The best model is Random Forest Regression.

The Random Forest Regression R^2 value (criterion= friedman_mse, maxFeatures= Sqrt & N_Estimators = 100) 0.8701340787917203

1. Multiple Linear Regression

R^2 value =0.7894790349867009

2. Support Vector Machine

S.No	Hyper Parameter	Linear (r value)	RBF (Non Linear) (r value)	Poly (r value)	SIGMOID (r value)
1	C10	-0.0064	-0.0807	-0.0935	-0.0906
2	C100	0.5412	-0.1228	-0.0979	-0.1179
3	C1	-0.1165	0.0883	-0.0639	-0.0899
4	C4	-0.0791	-0.0857	0.0828	-0.0900
5	C6	-0.0545	-0.0847	-0.0927	-0.0900
6	C8	0.0305	-0.0829	-0.0941	-0.0903

The SVN regression use R^2 value linear (r) and hyper parameter (C100) = 0.5412

3. DECISION Tree

S.No	Criterion	Max Features	Splitter	R value
1	friedman_mse	sqrt	best	0.7135736243878198
2	friedman_mse	sqrt	random	0.6373579770414344
3	friedman_mse	log2	best	0.7219566058751068
4	friedman_mse	log2	random	0.6480779798732756
5	friedman_mse	None	best	0.712964168702879
6	friedman_mse	None	random	0.6830959412444478
7	poisson	sqrt	best	0.5311500986191414
8	poisson	sqrt	random	0.6489984419374695
9	poisson	log2	best	0.70122401738235

10	poisson	log2	random	0.5781118618526647
11	poisson	None	best	0.7385762279418197
12	poisson	None	random	0.6917814151266277
13	absolute_error	sqrt	best	0.6669470691204613
14	absolute_error	sqrt	random	0.7111572289037678
15	absolute_error	log2	best	0.7525564583412355
16	absolute_error	log2	random	0.701272475261903
17	absolute_error	None	best	0.666427178789598
18	absolute_error	None	random	0.7583044079190236

The Decision Tree Regression use R^2 value (criterion= absolute_error , maxFeatures=None & splitter= random) 0.7583044079190236

4. Random Forest

S.No	Criterion	Max Features	N_Estimators	R value
1	friedman_mse	None	10	0.8435142329434163
2	friedman_mse	None	100	0.8515925732312891
3	friedman_mse	Sqrt	10	0.8521123994687568
4	friedman_mse	Sqrt	100	0.8701340787917203
5	friedman_mse	Log2	10	0.8533176475068392
6	friedman_mse	Log2	100	0.8694595846590039
7	poisson	None	10	0.8296240653657385
8	poisson	None	100	0.8545625662391444
9	poisson	Sqrt	10	0.8497943186680614
10	poisson	Sqrt	100	0.8692734942067537
11	poisson	Log2	10	0.8532871663931022
12	poisson	Log2	100	0.8678598047737017

The Random Forest Regression R^2 value (criterion= friedman_mse, maxFeatures= Sqrt & N_Estimators = 100) 0.8701340787917203